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An efficient edge computing management mechanism for sustainable smart cities

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ABSTRACT

Since ancient times, humanity has envisioned a world where people and things connect and interact. The advent of generation mobile systems, the so-called 5 G in the early century, has realized this dream. 5 G can provide network services with extremely-high throughput and extremely-low delay and allows a huge device number to connect based on Internet infrastructure, forming the Internet of Things (IoT). In recent years, IoT has been applied in various fields serving humanity, such as smart cities, smart agriculture, healthcare, education, military, and IoT ecosystems. One of the main challenges of IoT applications is computing solutions to reduce service response times. This study proposes an efficient edge computing management mechanism for IoT applications in smart cities. Our mechanism proposes a small database (called an information map) that allows edge computing servers of smart cities to store edge service information. When the mobile end-users move to the new edge server' managed coverage, properties related to the EC service are exchanged between the edge servers. The experiment results have shown that our proposed mechanism improves service response time and energy consumption many times compared to the traditional mechanism. We hope that this mechanism will be strongly applied to sustainable smart cities in the future.

1. Introduction

Mobile smart devices are becoming more popular and are necessary tools in modern society. The network connecting mobile devices is increasingly complex and diverse in types and technologies. Generation mobile systems, also known as 5 G, are topical and attractive research topics in both industry and academia [1,2]. The 5 G network delivers services with ultra-high throughput and ultra-low latency and allows hundreds of billions of mobile devices to connect. 5 G realizes the Internet of Things concept and forms a series of emerging concepts to serve humanity, such as smart cities, smart agriculture, smart healthcare, and other smart ecosystems [3].

According to Cisco's forecast, the number of connected IoT devices to the Internet will be over 500 billion by 2030 [4]. As a result, these devices will generate huge amounts of data. However, traditional data processing models and methods are infeasible [5]. For recent decades, cloud computing (CC) has been the dominant technology in all areas. CC has proven outstanding capability and ability by providing highly reliable computing services. However, one limitation of CC is high service response time, so it is unresponsive to real-time IoT applications [6]. To solve this problem, edge computing (EC) was proposed. EC brings the capabilities of the cloud closer to end-users, reducing service response times and realizing real-time IoT applications [7–9], as shown in Fig. 1.

Recently, the topics of smart cities based on IoT applications have been focused on research. Smart cities provide a wide range of convenient services for citizens, including smart energy management, smart healthcare, intelligent transportation systems, and other utility services. Therefore, it can be said that smart cities are the integration of Internet infrastructure, IoT technologies, and big data processing to provide citizens with increasingly superior smart utilities [10,11].

Our main contributions to the study can be summarized as follows.

– Determine 06 sub-components of smart cities, the concept of *smart people, smart economy, smart governance, smart mobility, smart environment, and, smart living.*

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– Perform a survey of proposed solutions and technologies to emphasize a vision of smart cities still has many challenges. The proposals are still in the primitive stage and are limited to each specific sector. More research and proposals with integrated solutions need to be employed for smart cities.
– Proposes an efficient edge computing management mechanism for smart cities to improve system performance, and energy consumption and realize real-time applications for smart cities.

The rest of the paper is presented as follows: The motivation to perform this study is presented in Section 2. In Section 3, we perform a short survey of recent proposals for smart cities. In Section 4, we present the proposed mechanism. Section 5 illustrates the effectiveness of the proposed mechanism, and Section 6 is the conclusion and proposes further research directions.

2. Motivation

We consider an EC-based IoT system for smart cities. This system connects IoT devices, sensors, actuators, applications, and citizens based on Internet infrastructure to provide convenient and intelligent utility services for residents, as described in Section 3. Specifically, smart IoT devices collect data. According to the traditional model, these data can be processed by different computing layers such as FC, EC, and CC [34–36] based on network communications within their coverage. Consequently, the data is sent to the EC centers, as shown in Fig. 2. However, in smart cities, end-users are often mobile. This leads to disconnection to the EC center when end-users move out of the EC center's coverage. Then the end-user will have to establish a new connection to another EC center and re-perform its service requests.

It should be noted that when an end-user move from one coverage to another, it needs to use a handover scheme. Recently, many efficient handover schemes have been proposed. In [37], A. Islam et al. proposed a secure non-orthogonal multiple access (NOMA) handover scheme for end-users based on blockchain. In [38], Kim et al. proposed a security handover schema between end-users and RSUs in vehicular networks. These handover schemes have demonstrated that their achieved high performance. In order to focus on highlighting the advantages of the proposed mechanism, we use the above existing handover schemes and ignore the constraints of the underlayers.

When an end-user connects to a new EC center, the results and services running at the old EC center are discarded and re-executed at the new EC center. As a result, service response time and energy and system resource consumption will increase. With hundreds of millions of mobile users in each city, we can imagine the huge resource and energy consumption. This is the major motivation to us perform this research.

3. Literature review

In recent, smart cities are one the emerging areas that have attracted great attention from the research community in both industry and academia. However, the concept of smart cities is very diverse. According to [12], the authors divide it into 06 sub-components, including *smart people* (citizens), *smart economy*, *smart governance*, *smart mobility* (smart transportation), *smart environment* (sustainable resource management), *smart living* (smart building, quality of life). Aiming to provide a comprehensive and in-depth vision of the research domain of smart cities, we perform a short survey of proposals in the period [2020–2022] in this domain. We divided the studies into six sub-components, as shown in Fig. 3. Besides we have also presented the applied technologies, objectives, and results of each study.

The survey results prove that the smart cities area is an exciting and topical research topic. In the last three years [2020–2022], the related studies have been published over 8.228 published on reputed digital library databases of IEEE Xplore, IET, River Publishers, MIT Press, and Wiley. The studies focus on some of the major solutions and technologies to develop sustainable and smart cities and improve the living quality of residents, such as IoT (1768), edge computing (935), AI (967), smart

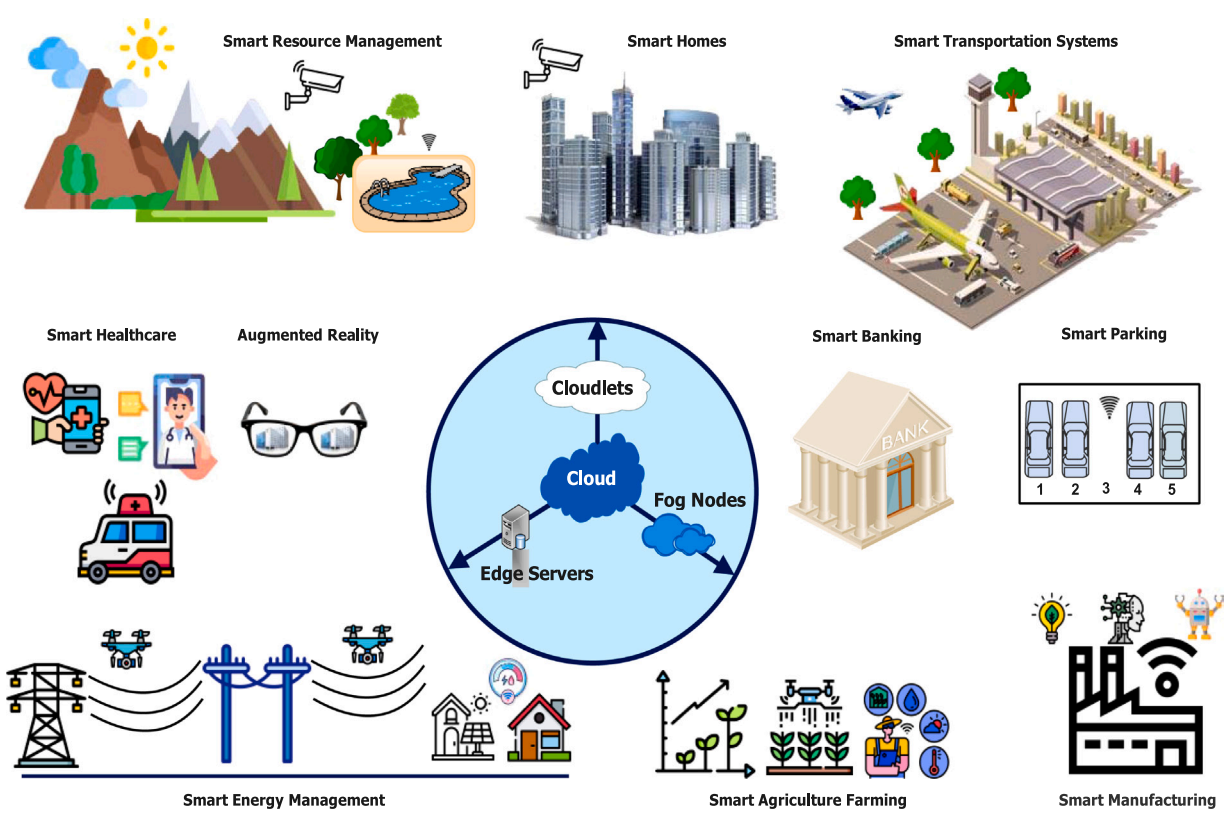

Fig. 1. Overview of EC-based IoT for smart cities.

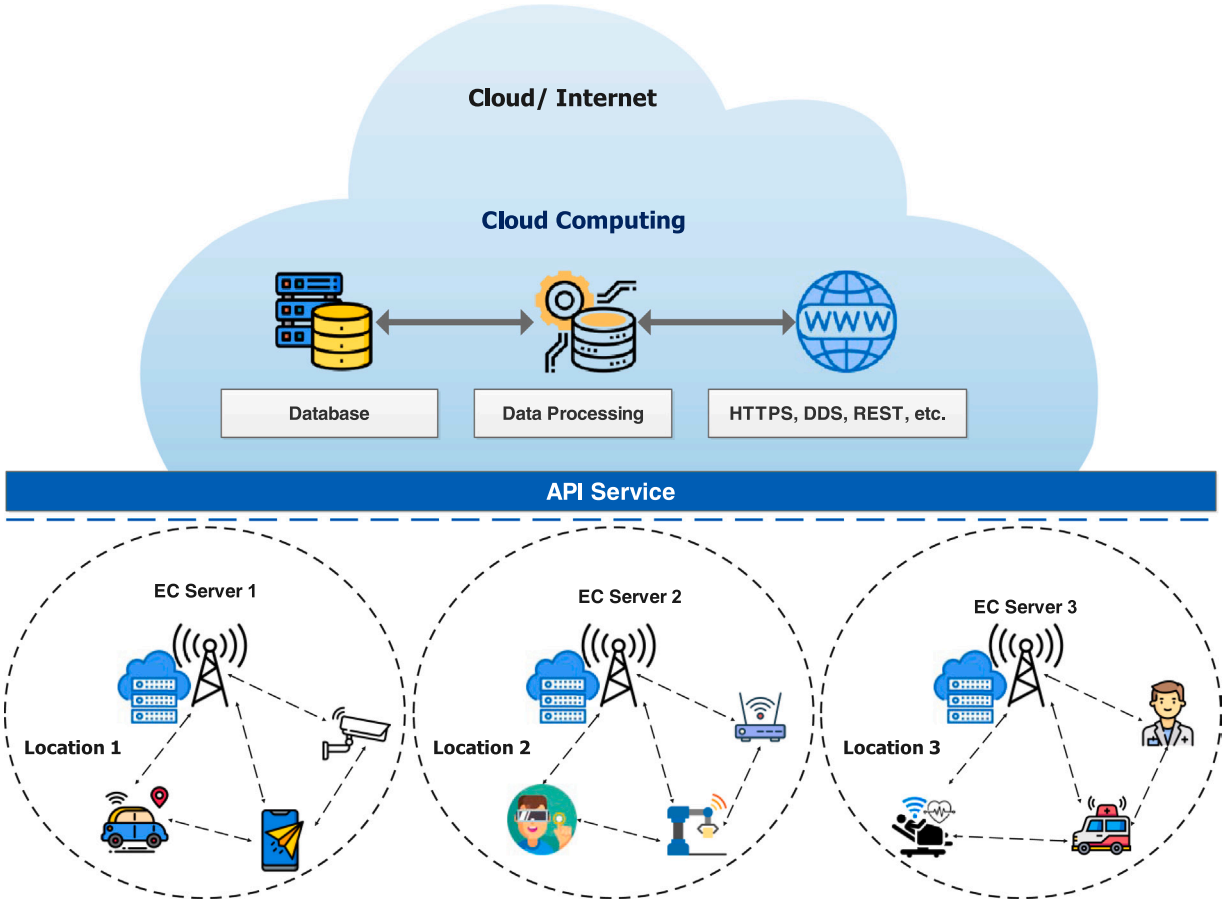

Fig. 2. The EC-based IoT architecture for smart cities.

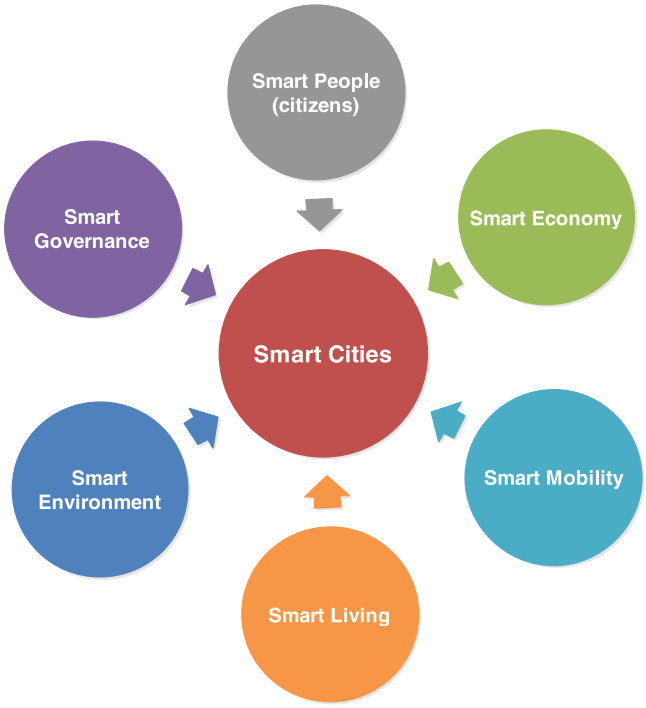

Fig. 3. The key components of smart cities.

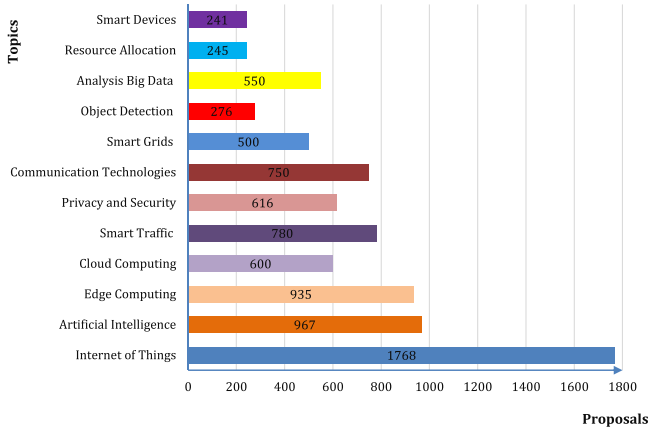

Fig. 4. Statistics on the number of proposed studies by topics in the period [2020–2022].

traffic (780), cloud computing (600), computing technologies (809), smart grid (500) and big data (550), as presented in Fig. 4. Analysis of some typical presented results are summarized in Table 1.

– *In the smart people sector, proposals focus on aspects to provide a more comfortable living for humans. Specifically, the authors have proposed a framework to enhance security and privacy* [13], accuracy in biometrics [14] and statistics, and analyze residents' sports activities [15].
– *In the smart economy sector, the solutions focus on aspects to develop a high and sustainable efficient economy for smart cities, including*

Table 1
The survey of recent proposed studies of Smart cities in period [2020–2022].

Ref.	Sectors	Technologies	Objective	Results
[13]	Smart people	IoT, Machine Learning, Blockchain	Design a data privacy and security framework for IoT-based smart cities	Use a 2-level privacy method to enhance the accurate detection ratio of attacks by 91 % and improve performance
[14]		5 G, CNN, Deep Learning	Design a biometric system to enhance the accuracy detection ratio of the alteration and fake biometric issues in smart cities.	Improve the detection of fingerprint alterations with superior performance up to 92.97 % and 98.15 % with CCN and ConvLSTM models, respectively
[15]		Deep Learning, CNN, Smart devices	Propose a statistical-deep model rely on wearable sensor devices to recognize citizen's physical activities in smart cities	Improve recognition performance with accuracy up to 97.4 % on three public data sets, including Daily, Sports and RealWorld activities
[16]	Smart economy	Bigdata, IoT	Propose an IoT-based smart seaport model for the sustainable economic development of smart cities	This research claimed that IoT-based seaports have huge potential to reduce cost and energy savings, enhance accuracy in seaport activities, and economic development of smart cities
[17]		EV2X, Queuing Model	Propose a charging slot booking trading framework for electric vehicles in smart cities	Improve accuracy, and saving of charging time and costs by up to 40 % and 93.3 %, respectively
[18]		IoV, Edge Computing, Blockchain	Propose a paid EC resource allocation trading system rely on IoV for smart cities	Enhance performance, security, and stability of the system and optimize the utilities, and pricing to encourage partners to join the system
[19]	Smart Mobility	Internet of EVs (IoEVs)	Propose the load-balancing wireless charging allocation algorithm for electric vehicles in smart cities	Improve PAER parameter up to 44 % and increase the double trip length on DWC lanes to compensate for consumed energy
[20]		Deep Learning	Propose a deep learning-based travel time estimation model by using neighbor's historical data	Improve the accuracy of the estimation time compared to existing models for smart traffic applications in smart cities
[21]		IoV, Heuristic algorithms	Propose a real-time heuristic-based optimization algorithm for cost efficiency of the IoV in smart cities	Improve up to 11 % cost with moderate delay around 0.8 s and number of switched RSUs reduce to 15 %.
[22]	Smart Environment	Google Map, GPS, C-Means Clustering	Propose an optimal path-finding algorithm base on parameters of traffic congestion and air quality for smart cities	Improve performance in terms of the path distance, air quality, and congestion status compared to existing algorithms using the Tehran city map
[23]		LoRa, WiFi, Cluster algorithms	Design a communication protocol for IoT-based irrigation systems in smart cities	Improve system performance compared to existing protocols and operates well for both WiFi and LoRa devices
[24]		AI, IoT, evolutionary algorithms	Propose an IoT-based waste management model for smart environment in smart cities	Improve waste management processes to improve the quality of life in smart cities of India.
[25]	Smart Living	IoT, AI, 5 G	Design Point of Interest recommendation system for smart tourism based on AI and 5 G-IoT	Improve system performance in terms of data rate and precision compared to existing systems
[26]		AI, IoT	Propose a management and recognition solution based on AI to evaluate the crowd's behaviors in smart cities	Improve privacy, security and performance in terms of control, and monitoring of crowds compared to existing systems.
[27]		IoT, Cloudlet,	Propose an IoT-based e-Health data management framework for smart cities to enhance citizen's living quality	Improve data privacy and security, service response time, reduces energy consumption of IoT devices by 50 % of IoT devices compared to existing cloud-based solutions.
[28]		AI, Drones and Blockchain	Propose a secure data aggregation scheme based on fusing FL, Blockchain for Drones-assisted IoT systems.	Improve significantly on aspects of data transmission rate, execution time, and mean absolute error compared to existing schemes.

(*continued on next page*)

Table 1 (*continued*)

Ref.	Sectors	Technologies	Objective	Results
[29]	Smart Governance	IoT, Blockchain, Game Theory	Propose a public participation blockchain architecture for smart governance of the infrastructure maintenance in smart cities	Improve security and encourage the public to participate in the governance process of smart cities.
[30]		IoT, Software-Defined Networking	Propose an SDN-based complex event processing framework for governance in smart cities	Improve in terms of energy consumption and system performance traffic, service response time compare to existing solutions
[31]		IoT, 5 G, Blockchain	Design a blockchain-based e-voting solution for governance in smart cities	Enhance security, against DoS, DDoS attacks, improve accuracy ratio and authentication delay

improving the efficiency of seaport operations [16], charging-booking management solutions for electric vehicles [17], and paid mobile edge computing resource allocation solution [18].

– *In the smart mobility sector, the studies focus on solving the challenges of smart transportation systems, namely the radio charging resource allocation solutions for electric vehicles* [19]. Other works focus on identifying optimal route routing and travel time estimation [20], cost optimization, and service response time for IoV applications [21].

– *In the smart environments sector, the proposals focus on solutions to provide a green and friendly living environment for residents of smart cities. Specifically, the authors have proposed solutions to determine optimal travel on the aspects of air quality and costs* [22], smart irrigation solutions [23], and smart environmental management models [24].

– *In the smart living sector, studies focus on solutions to improve the life quality of residents, including smart travel recommendations* [25], crowd monitoring and management [26], smart healthcare [27], and crowdsourcing schemes based on emerging technologies consisting of Drones, AI, and Blockchain [28].

– *In the smart governance sector, proposals focus on proposing transparent and effective urban governance solutions, including infrastructure management with the participation and making-decision of residents* [29], complex event management mechanisms [30], and real-time electronic voting solutions [31].

The surveyed results have shown that the vision of smart cities still has many challenges. The proposals are still in the primitive stage and are limited to each specific sector. More research and proposals with integrated solutions need to be employed in this field. Studies have also demonstrated that the heart of smart cities is the citizens. Therefore, the awareness, applicability, and adoption of new technologies, as well as the active participation of residents, are major problems.

Moreover, a common one of the most challenging problems of these applications is the response time, special for real-time or delay-sensitive systems. To address this problem, recent studies have proposed edge computing (EC) solutions [32]. Instead of centralized data processing on

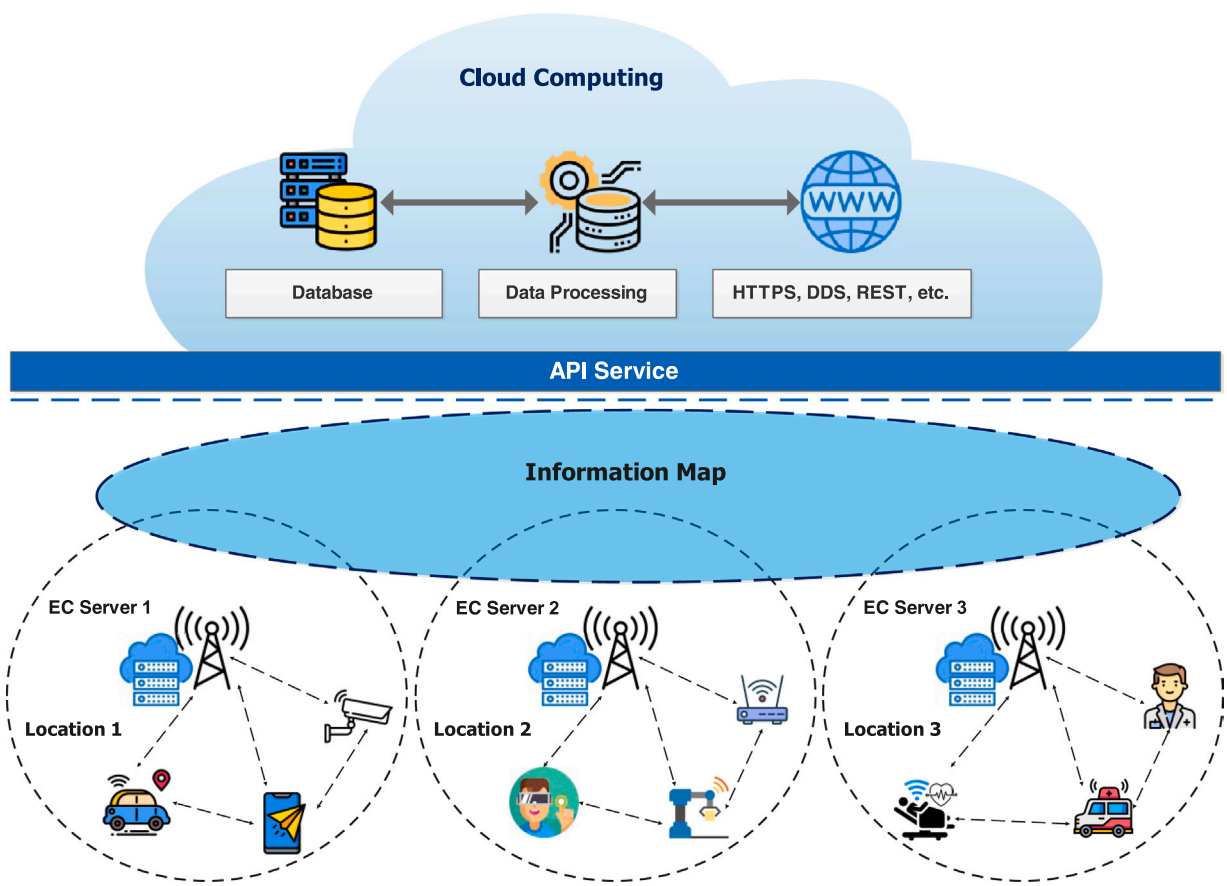

Fig. 5. The proposed EC-based IoT architecture for smart cities.

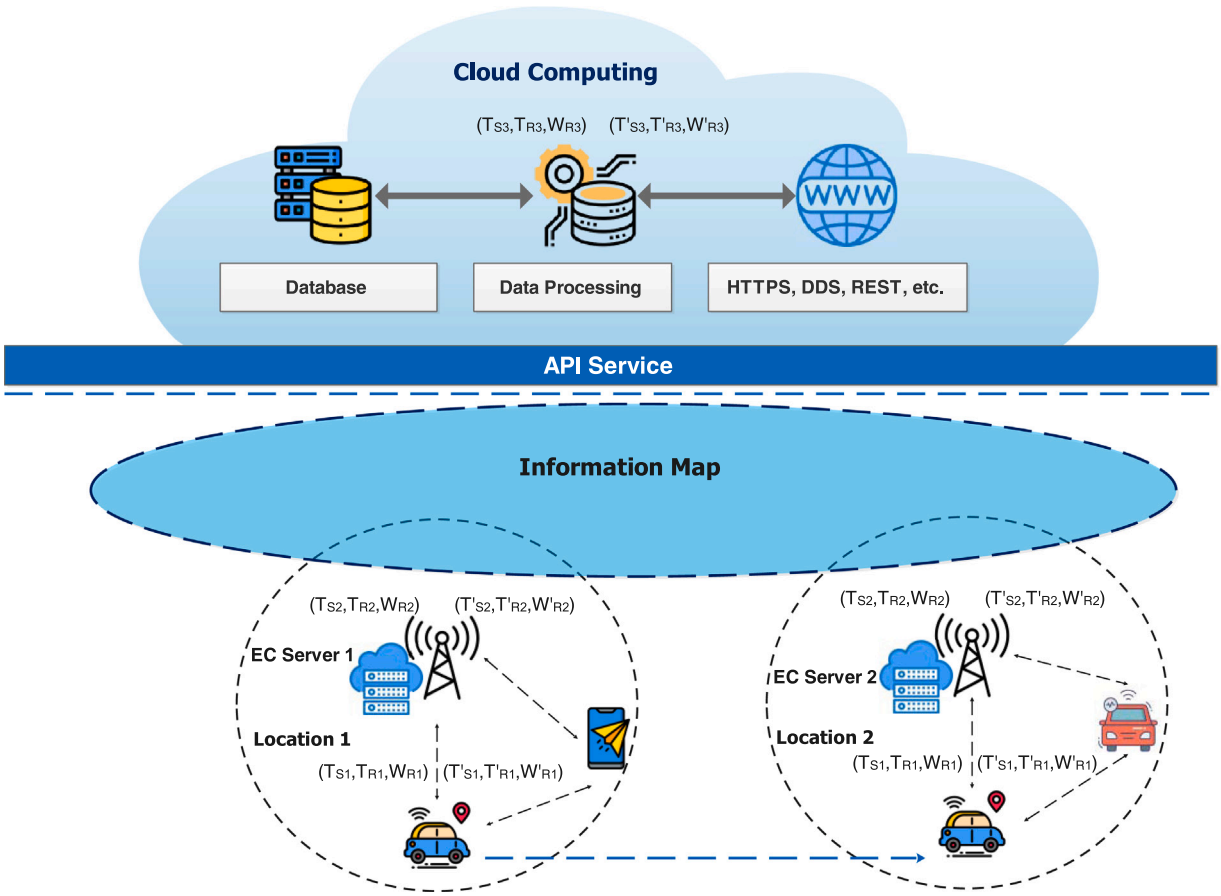

Fig. 6. An illustration of the working principle of the proposed mechanism.

the cloud, information is handled at the edge of networks [33]. As a result, it reduces response time, energy consumption, computing cost, and overall performance and realizes real-time applications. In this study, we propose an efficient edge servers management mechanism for real-time IoT applications in smart cities. Indeed, our mechanism allows edge servers to exchange information with each other such as EC properties and services that have been executed at edge servers through an information map. Thanks to these issues, our mechanism reduces significantly service response time, energy consumption, and resources.

In our opinion, although there are still many challenges, smart cities will be an inevitable development trend in the future. The detailed issues will be presented in the following sections.

4. Proposed mechanism

4.1. Network architecture

In [39], Wang et al. introduced the architecture of EC-based IoT applications for smart cities. This architecture consists of three layers (Fig. 2), including:

- *Things Layer:* This layer includes devices, sensors, smart IoT, and actuators to collect and provide data to the upper layers as well as execute service requests from the upper layers.
- *Edge Layer:* This layer includes servers that provide EC services (EC centers) to meet the service requirements of the Things layer and play an intermediary role in communications between the cloud layer and the Things layer.
- *Cloud Layer:* This layer includes robust servers with high computing, huge storage capacity, and extremely high connection bandwidth to provide the ability to store, process, and compute big data or handle problems beyond the responsiveness of the EC layer.

It should be recalled that the two basic transactions in a computer network environment are searching and routing [40]. These transactions are invoked to serve service requests derived from the things layer with objectives including minimal service response time, saving energy, and system resource consumption. To solve these problems, we propose an information map between the EC centers in the proposed architecture, as shown in Fig. 5.

The information map is defined as follows: An information map is a small database to store properties and services that the EC servers have provided. When an end-user moves from one coverage to another, instead of it launching a search progressing on the cloud to fulfill service requirements that were previously served or EC servers executed. Now, EC servers only perform searching within the information map.

Indeed, when end-user moves between coverages, they are controlled by advanced handover schemes [37,38], as described in Section 2. The handover process is seamless. Another point from our mechanism is the EC services, which have been executed in the past, will be stored in the information map. At the next point, end-users can immediately access these services, without initiating a search process in the cloud. Besides, the information map is stored on edge servers and takes up a portion of the resources of edge servers. The size of the information map is self-established and self-configured based on the abilities and capabilities of the edge servers through the intelligence caching schemes at the edge such as the capacity-aware [41], version-aware caching schemes [42].

Although maintaining an information map will consume a certain amount of energy and resources. In our opinion, it is negligible

Table 2
The mathematical symbols and meaning.

Order	Parameters	Meaning
1	T_{S1}	The total time cost to perform searching transactions at the Things layer
2	T_{S2}	The total time cost to perform searching transactions at the edge layer
3	T_{S3}	The total time cost to perform searching transactions at the cloud layer
4	T_{R1}	The total time cost to perform routing transactions at the Things layer
5	T_{R2}	The total time cost to perform routing transactions at the edge layer
6	T_{R3}	The total time cost to perform routing transactions at the cloud layer
7	T'_{R1}	The total time cost to reply routing transactions at the Things layer
8	T'_{R2}	The total time cost to reply routing transactions at the edge layer
9	T'_{R3}	The total time cost to reply routing transactions at the cloud layer
10	W_{S1}	The amount of energy consumption for a searching transaction at the Things layer
11	W_{S2}	The amount of energy consumption for a searching transaction at the edge layer
12	W_{S3}	The amount of energy consumption for a searching transaction at the cloud layer
13	W_{R1}	The amount of energy consumption for a routing transaction at the Things layer
14	W_{R2}	The amount of energy consumption for a routing transaction at the Things layer
15	W_{R3}	The amount of energy consumption for a routing transaction at the Things layer
16	W'_{R1}	The amount of energy consumption to reply results in the Things layer
17	W'_{R2}	The amount of energy consumption to reply results at the edge layer
18	W'_{R3}	The amount of energy consumption to reply results at the edge layer
19	$T_{Tranditional}$	The time cost to perform searching and routing transactions per service request according to the traditional solution
20	$T_{Proposed}$	The time cost to perform searching and routing transactions per request according to the proposed solution
21	$W_{Tranditional}$	The amount of energy consumption to perform per request according to the traditional solution
22	$W_{Proposed}$	The amount of energy consumption to perform one request according to the proposed solution

compared to the resource and energy consumption of the cloud as well as the trade-off cost to pay for achieving minimal response time. Obviously, searching transactions on information maps helps to reduce service response time, computing costs, energy, and resource consumption many times compared to the existing normal solutions.

Thanks to the information map, the proposed solution improves robustly the overall system performance, and computing costs and realizes real-time applications. The proposed architecture is shown in Fig. 5, and the working principle of the proposed mechanism is presented in Fig. 6.

4.2. Efficient analysis of mechanism

Aims to the convenience of the reader, we present some mathematical symbols and their meanings as follows:

- Symbol n is the number of roaming service requests while it is in transaction progress.
- Symbols T and T', respectively, are the service request time and service response time at each layer and are the same for all coverage.
- Symbols W and W', respectively, are the amount of consumed energy of a service request and a service response, respectively, and are equal for all coverage.

Table 3
The simulation parameters and values.

Order	Parameters	Values
1	T_{S1}	$Random[1-5]$
2	T_{S2}	$Random[1-T_{S1}]$
3	T_{S3}	$Random[5-10]$
4	T_{R1}	$Random\ of[1-3]$
5	T_{R2}	$Random[1-T_{R1}]$
6	T_{R3}	$Random[1-3]$
7	T'_{R1}	$Random[1-T_{R1}]$
8	T'_{R2}	$Random[1-T_{R1}]$
9	T'_{R3}	$Random[1-3]$
10	W_{S1}	$Random[1-3]$
11	W_{S2}	$Random[W_{S1}-5]$
12	W_{S3}	Random $[W_{S2}$-10]
13	W_{R1}	$Random[1-2]$
14	W_{R2}	$Random[1-2]$
15	W_{R3}	$Random[1-2]$
16	W'_{R1}	$Random[1-2]$
17	W'_{R2}	$Random[1-2]$
18	W'_{R3}	$Random[1-2]$

Assume that the Things, edge and cloud layers, respectively, of each cluster consume the amount of energy is the same per time unit. Some remaining mathematical symbols and their meanings are summarized in Table 2.

According to the proposed operating model in Fig. 6, we have established the calculating equations of time cost and energy consumption of the system as follows.

$$T_1 = T_2 = \sum_{i=1}^{n} (T_{S1} + T_{S2} + T_{S3} + T_{R1} + T_{R2} + T_{R3}) \tag{1}$$

$$T'_1 = \sum_{i=1}^{n} (T'_{R1} + T'_{R2} + T'_{R3}) \tag{2}$$

$$W_1 = W_2 = \sum_{i=1}^{n} (W_{S1} + W_{S2} + W_{S3}) + \sum_{i=1}^{n} (W_{R1} + W_{R2} + W_{R3}) \tag{3}$$

$$W'_1 = W'_2 = \sum_{i=1}^{n} (W'_{R1} + W'_{R2} + W'_{R3}) \tag{4}$$

In case the mobile node only moves within coverage, the service response time and energy consumption could be determined by Eqs. (5) and (6), as follows.

$$T = T_1 + T'_1 \tag{5}$$

$$W = W_1 + W'_1 \tag{6}$$

Suppose that after a mobile user sends a service request to the higher layers, and then it moves to new coverage. In this case, the traditional mechanism will perform searching and routing transactions on both coverages. Consequently, time cost and energy consumption could be determined by Eqs. (7) and (8) as follows.

$$T = T_1 + T_2 + T'_2 \tag{7}$$

$$W = W_1 + W'_1 + W_2 + W'_2 \tag{8}$$

In our proposed model, the service response time and energy consumption are still determined as in Eqs. (7) and (8), respectively, for the old cluster. However, when the mobile node moves to new coverage, due to searching results have been stored in the information map, this mobile user only performs the searching transactions on the information map but not queries to the cloud. Consequently, T_{S2}, T'_{S2}, W_2, W'_2 values are redefined according to the Eqs. (10)–(15), as follows.

$$T_2 = \sum_{i=1}^{n}(T_{S1} + T_{S2}) + \sum_{i=1}^{n}(T_{R1} + T_{R2}) \tag{9}$$

$$T_2^{'} = \sum_{i=1}^{n}(T_{R1}^{'} + T_{R2}^{'}) \tag{10}$$

$$W_2 = \sum_{i=1}^{n}(W_{S1} + W_{S2}) + \sum_{i=1}^{n}(W_{R1} + W_{R2}) \tag{11}$$

$$W_2^{'} = \sum_{i=1}^{n}(W_{R1}^{'} + W_{R2}^{'}) \tag{12}$$

- *Traditional Mechanism:* apply Eqs. (1)–(8) to determine the spend time and energy consumption of the system
- *Proposed Mechanism*: T_1, T'_1, W_1, W'_1 values apply Eqs. (1)–(4), however, T_2, T'_2, W_2, W'_2 values apply Eqs. (9)–(12).

As a result, the search time and energy consumption are often reduced and proportional to the number of roaming service requests. In the next section, we perform a simple experiment to highlight the efficiency of the proposed solution.

5. Performance evaluation

In this section, we set up experiments to evaluate the performance of the proposed solution compared to traditional mechanisms (refer to [43] for more detail). Owning to the moving of IoT devices, the provided network services can be disconnected. The re-establishing will motivate searching and routing transactions. By application information map, edge servers can leverage executed EC services on other edge servers to quickly direct the request to the suitable servers. The settings, metrics and results, and analyzes are presented as follows.

5.1. Simulation settings

Aiming to clarify the effectiveness of the proposed mechanism, we set up an EC-based IoT system consisting of $N = 1000$ mobile end-user nodes, the nodes are evenly distributed in clusters, and each cluster is a coverage. We have assumed that the maximum percentage ($\lambda_{\max}$) of nodes roaming in while the node's service requests are becoming processed by 25 %, which is equivalent to the maximum number of roaming mobile nodes (symbol n), $n = Random[1 - 250]$. Moreover, to clarify the mechanism's efficiency compared to the existing traditional mechanisms (refer to [43] for details), we have also assumed that the costs of routing and searching transactions of different coverage are the same and represented by natural numbers through a random function. As a result, we obtain the parameter values as presented in Table 3.

5.2. Metrics

In this section, we present how to determine and calculate simulation results. Indeed, we have shown the efficiency of the proposed mechanism in two aspects, time cost, and energy consumption. For each of these aspects, we consider efficiency on a per mobile device or the whole system. We define how to calculate the simulation results as follows.

– Let $T_{Tranditional}$ and $T_{Proposed}$, are time cost per each mobile device of the traditional method and the proposed method, respectively, as follows:

$$T_{Tranditional} = 2 \times \sum_{i-1}^{n}(T_{S1} + T_{S2} + T_{S3} + T_{R1} + T_{R2} + T_{R3}) + \sum_{i-1}^{n}(T'_{R1} + T'_{R2} + T'_{R3}) \tag{13}$$

$$T_{Proposed} = \sum_{i-1}^{n}(T_{S1} + T_{S2} + T_{S3} + T_{R1} + T_{R2} + T_{R3}) + \sum_{i-1}^{n}(T_{S1} + T_{S2} + T_{R1} + T_{R2}) + \sum_{i-1}^{n}(T'_{R1} + T'_{R2}) \tag{14}$$

– Let $T_{Total-Tranditional}$ and $T_{Total-Tranditional}$, are the total of time cost of whole system of the traditional method and the proposed method, respectively, as follows:

$$T_{Total-Tranditional} = n \times T_{Tranditional} \tag{15}$$

$$T_{Total-Tranditional} = n \times T_{Proposed} \tag{16}$$

– Let $W_{Tranditional}$ and $W_{Proposed}$, are the energy consumption per each mobile device of the traditional method and the proposed method, respectively, as follows:

$$W_{Tranditional} = 2 \times \sum_{i-1}^{n}(W_{S1} + W_{S2} + W_{S3} + W_{R1} + W_{R2} + W_{R3}) + \sum_{i-1}^{n}(W'_{R1} + W'_{R2} + W'_{R3}) \tag{17}$$

$$W_{Proposed} = \sum_{i-1}^{n}(W_{S1} + W_{S2} + W_{S3} + W_{R1} + W_{R2} + W_{R3}) + \sum_{i-1}^{n}(W_{S1} + W_{S2} + W_{R1} + W_{R2}) + \sum_{i-1}^{n}(W'_{R1} + W'_{R2}) \tag{18}$$

– Let $E_{Total-Tranditional}$ and $E_{Total-Proposed}$, are the total energy consumption of whole system of the tranditional method and the proposed method, respectively, as follows:

$$E_{Total-Tranditional} = n \times W_{Tranditional} \tag{19}$$

$$E_{Total-Proposed} = n \times W_{Proposed} \tag{20}$$

5.3. Results and analysis

The experimental results in Figs. 7–10 have presented the efficiency of the proposed mechanism compared to the traditional mechanism on aspects of service response time and energy consumption, respectively.

Fig. 7 presents a performance comparison of two mechanisms in terms of the average service response time per service request at the things layer. The simulation results show that the proposed mechanism reduces the service response time compared to the traditional mechanism.

Fig. 8 presents a performance comparison of aspects of the service response time of the whole system. The results have shown that our mechanism has significantly reduced the service response time of all requests from mobile end-users compared to the traditional mechanism. Furthermore, the effectiveness of the proposed mechanism is higher as the number of roaming service requests increases, proportional to the increase in the number of mobile end-users.

On the energy consumption hand, Fig. 9 presents a performance comparison of two mechanisms in terms of energy consumption per service request at the Things layer. The simulation results have shown that our mechanism has improved energy consumption by [15 − 30] % compared to the traditional mechanism.

Fig. 10 has also shown the total energy consumption of the whole system improved significantly, up to 40 %, compared to the conventional scheme in some cases.

The simulation results have demonstrated that thanks to the addition of information maps, service response times and energy consumption are robustly improved compared to existing traditional solutions. In real smart cities, edge servers of smart transportation systems, and crowdsourcing systems can apply this solution by establishing information maps between edge servers to optimize services and enhance user

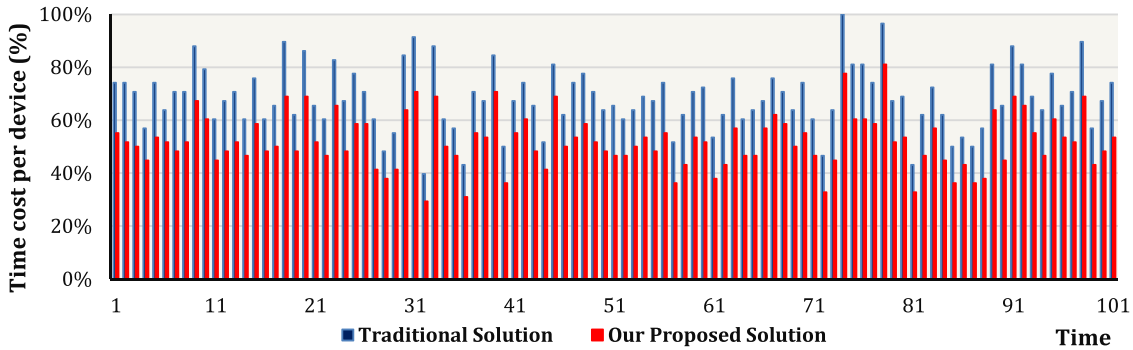

Fig. 7. The efficiency of 2-mechanisms in terms of the service response time on each device.

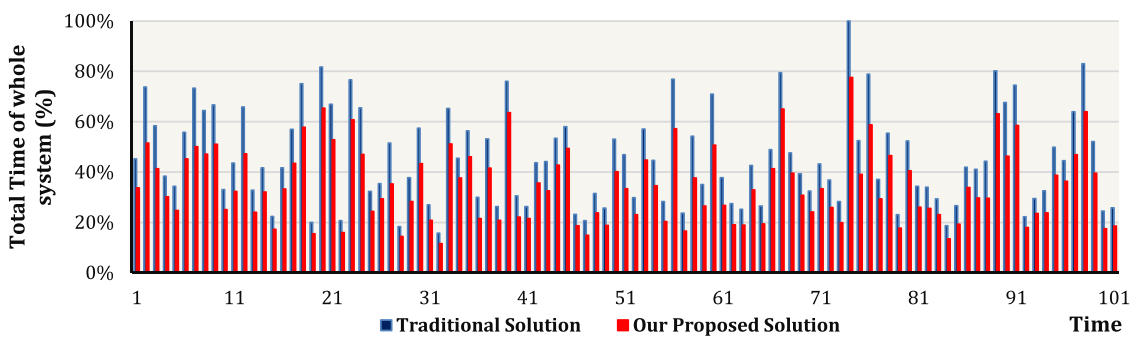

Fig. 8. The efficiency of 2-mechanisms in terms of the service response time on the whole system.

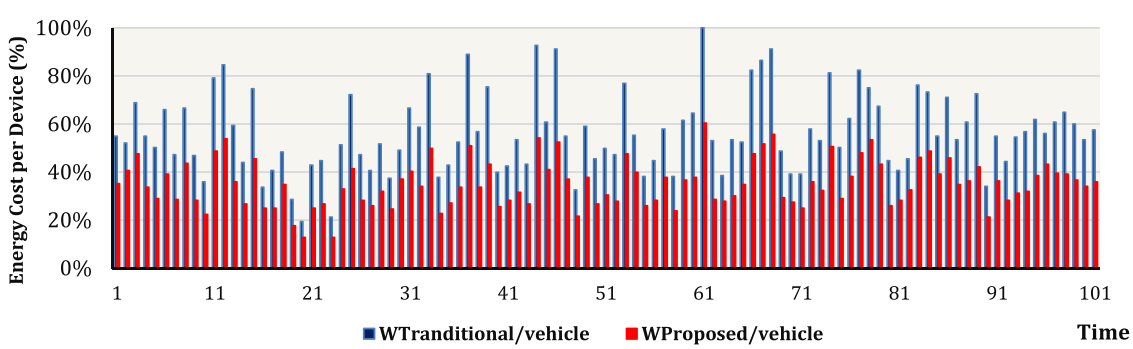

Fig. 9. The efficiency of 2-mechanisms in terms of the energy consumption on each device.

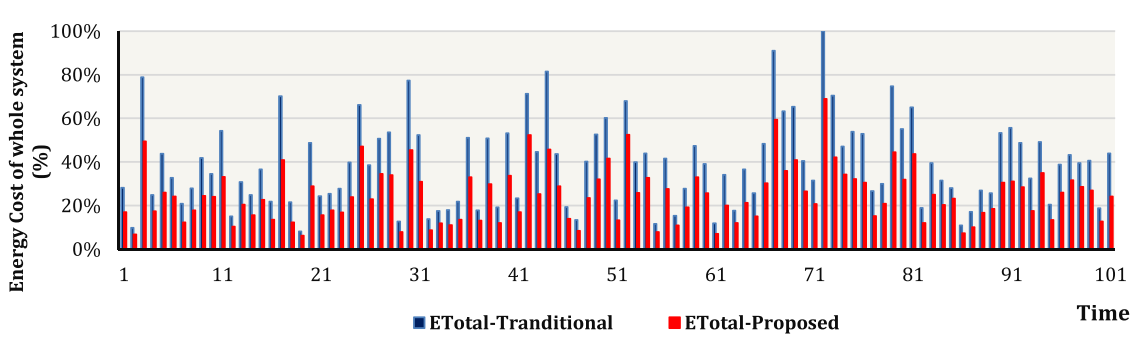

Fig. 10. The efficiency of 2-mechanisms in terms of the energy consumption on the whole system.

experiences.

6. Conclusion and open issues

This study focuses on researching and proposing a management mechanism between EC servers to reduce service response time, energy efficiency, and system resources for EC-based IoT systems in smart cities. Our proposed mechanism works at the application layer between EC servers by establishing an information map to store properties of EC services that the executed EC servers. This mechanism is highly feasible and very effective in reducing service response time, saving energy, and improving the overall performance of IoT systems for smart cities.

Calculations have demonstrated that our solution outperforms compared to traditional computational solutions. A limitation that needs further research is when the information map is too large, it will consume more energy to store the map. In our opinion, storing this information map requires cloud participation through establishing a mechanism for information exchange between EC and CC. Moreover, the sharing of information maps should be combined with state-of-the-art secret share techniques [44,45] against attacks from the increasing complexity of network environments. These issues will be studied in the future time. We hope that, with its simplicity but high practicability, our solution will be strongly applied to cities more become smarter and more sustainable for humanity.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

References

[1] R. Khan, P. Kumar, D.N.K. Jayakody, M. Liyanage, A survey on security and privacy of 5G technologies: potential solutions, recent advancements, and future directions, IEEE Commun. Surv. Tutor. 22 (1) (2020) 196–248, https://doi.org/10.1109/COMST.2019.2933899.

[2] Y. Siriwardhana, P. Porambage, M. Liyanage, M. Ylianttila, A survey on mobile augmented reality with 5G mobile edge computing: architectures, applications, and technical aspects, IEEE Commun. Surv. Tutor. 23 (2) (2021) 1160–1192, https://doi.org/10.1109/COMST.2021.3061981.

[3] S. Wijethilaka, M. Liyanage, Survey on network slicing for internet of things realization in 5G networks, IEEE Commun. Surv. Tutor. 23 (2) (2021) 957–994, https://doi.org/10.1109/COMST.2021.3067807.

[4] Y.B. Zikria, R. Ali, M.K. Afzal, S.W. Kim, Next-generation internet of things (IoT): opportunities, challenges, and solutions, Sensors 21 (2021) 1174, https://doi.org/10.3390/s21041174.

[5] X. Wang, Y. Han, V.C.M. Leung, D. Niyato, X. Yan, X. Chen, Convergence of edge computing and deep learning: a comprehensive survey, IEEE Commun. Surv. Tutor. 22 (2) (2020) 869–904, https://doi.org/10.1109/COMST.2020.2970550.

[6] H. Wang, et al., Architectural design alternatives based on cloud/edge/fog computing for connected vehicles, IEEE Commun. Surv. Tutor. 22 (4) (2020) 2349–2377, https://doi.org/10.1109/COMST.2020.3020854.

[7] H. El-Sayed, et al., Edge of things: the big picture on the integration of edge, IoT and the cloud in a distributed computing environment, IEEE Access 6 (2018) 1706–1717, https://doi.org/10.1109/ACCESS.2017.2780087.

[8] H. Qiu, K. Zhu, N.C. Luong, C. Yi, D. Niyato, D.I. Kim, Applications of auction and mechanism design in edge computing: a survey, IEEE Trans. Cogn. Commun. Netw. 8 (2) (2022) 1034–1058, https://doi.org/10.1109/TCCN.2022.3147196.

[9] W. Rafique, L. Qi, I. Yaqoob, M. Imran, R.U. Rasool, W. Dou, Complementing IoT services through software defined networking and edge computing: a comprehensive survey, IEEE Commun. Surv. Tutor. 22 (3) (2020) 1761–1804, https://doi.org/10.1109/COMST.2020.2997475.

[10] L.U. Khan, I. Yaqoob, N.H. Tran, S.M.A. Kazmi, T.N. Dang, C.S. Hong, Edge-computing-enabled smart cities: a comprehensive survey, IEEE Internet Things J. 7 (10) (2020) 10200–10232, https://doi.org/10.1109/JIOT.2020.2987070.

[11] D.G. Costa, et al., A survey of emergencies management systems in smart cities, IEEE Access 10 (2022) 61843–61872, https://doi.org/10.1109/ACCESS.2022.3180033.

[12] A. Kirimtat, O. Krejcar, A. Kertesz, M.F. Tasgetiren, Future trends and current state of smart city concepts: a survey, IEEE Access 8 (2020) 86448–86467, https://doi.org/10.1109/ACCESS.2020.2992441.

[13] Prabhat Kumar, Randhir Kumar, Gautam Srivastava, Govind P. Gupta, Rakesh Tripathi, Thippa Reddy Gadekallu, Neal N. Xiong, PPSF: A privacy-preserving and secure framework using blockchain-based machine-learning for IoT-driven smart cities, IEEE Trans. Netw. Sci. Eng. volume. 8 (2021) 2326–2341, https://doi.org/10.1109/TNSE.2021.3089435.

[14] Ahmed Sedik, Lo'Ai Tawalbeh, Mohamed Hammad, Ahmed A. Abd El-Latif, Ghada M. El-Banby, Ashraf A.M. Khalaf, Fathi E. Abd El-Samie, Abdullah M. Iliyasu, Deep learning modalities for biometric alteration detection in 5G networks-based secure smart cities, IEEE Access 9 (2021) 94780–94788, https://doi.org/10.1109/ACCESS.2021.3088341.

[15] Thien Huynh-The, Cam-Hao Hua, Nguyen Anh Tu, Dong-Seong Kim, Physical activity recognition with statistical-deep fusion model using multiple sensory data for smart health, IEEE Internet Things J. 8 (2020) 1533–1543, https://doi.org/10.1109/JIOT.2020.3013272.

[16] Kok-Lim Alvin Yau, Shuhong Peng, Junaid Qadir, Yeh-Ching Low, Mee Hong Ling, Towards smart port infrastructures: enhancing port activities using information and communications technology, IEEE Access 8 (2020) 83387–83404, https://doi.org/10.1109/ACCESS.2020.2990961.

[17] Suresh Chavhan, Nikhil Dubey, Abhinav Lal, Dev Khetan, Deepak Gupta, Ashish Khanna, Joel J.P.C. Rodrigues, Plácido Rogerio Pinheiro, Next-generation smart electric vehicles cyber physical system for charging slots booking in charging stations, IEEE Access 8 (2020) 160145–160157, https://doi.org/10.1109/ACCESS.2020.3020115.

[18] Xi Lin, Jun Wu, Shahid Mumtaz, Sahil Garg, Jianhua Li, Mohsen Guizani, Blockchain-based on-demand computing resource trading in IoV-assisted smart city, IEEE Trans. Emerg. Top. Comput. 8 (2021) 1373–1385, https://doi.org/10.1109/TETC.2020.2971831.

[19] Eiman ElGhanam, Hazem Sharf, Yazan Odeh, Mohamed S. Hassan, Ahmed H. Osman, On the coordination of charging demand of electric vehicles in a network of dynamic wireless charging systems, IEEE Access 10 (2022) 62879–62892, https://doi.org/10.1109/ACCESS.2022.3182700.

[20] Jing Qiu, Lei Du, Dongwen Zhang, Shen Su, Zhihong Tian, Nei-TTE: intelligent traffic time estimation based on fine-grained time derivation of road segments for smart city, IEEE Trans. Ind. Inform. 16 (2020) 2659–2666, https://doi.org/10.1109/TII.2019.2943906.

[21] Luca Cesarano, Andrea Croce, Leandro Do Carmo Martins, Daniele Tarchi, Angel A. Juan, A real-time energy-saving mechanism in internet of vehicles systems, IEEE Access 9 (2020) 157842–157858, https://doi.org/10.1109/ACCESS.2021.3130125.

[22] Elham Ghaffari, Amir Masoud Rahmani, Morteza Saberikamarposhti, Amir Sahafi, An optimal path-finding algorithm in smart cities by considering traffic congestion and air pollution, IEEE Access 10 (2022) 55126–55135, https://doi.org/10.1109/ACCESS.2022.3174598.

[23] Abdulaziz Aldegheishem, Nabil Alrajeh, Laura García, Jaime Lloret, SWAP: smart water protocol for the irrigation of urban gardens in smart cities, IEEE Access 10 (2022) 39239–39247, https://doi.org/10.1109/ACCESS.2022.3165579.

[24] Mohammadhossein Ghahramani, MengChu Zhou, Anna Molter, Francesco Pilla, IoT-based route recommendation for an intelligent waste management system, IEEE Internet Things J. 9 (2022) 11883–11892, https://doi.org/10.1109/JIOT.2021.3132126.

[25] Wei Wang, Neeraj Kumar, Junxin Chen, Zhiguo Gong, Xiangjie Kong, Wei Wei, Honghao Gao, Realizing the potential of the internet of things for smart tourism with 5G and AI, IEEE Netw. 34 (2020) 295–301, https://doi.org/10.1109/MNET.011.2000250.

[26] Juan Ramón Santana, Luis Sánchez, Pablo Sotres, Jorge Lanza, Tomás Llorente, Luis Muñoz, A privacy-aware crowd management system for smart cities and smart buildings, IEEE Access 8 (2020) 135394–135405, https://doi.org/10.1109/ACCESS.2020.3010609.

[27] Souvik Sengupta, Suman Sankar Bhunia, Secure data management in cloudlet assisted iot enabled e-health framework in smart city, IEEE Sens. J. 20 (2020) 9581–9588, https://doi.org/10.1109/JSEN.2020.2988723.

[28] A. Islam, A. Al Amin, S.Y. Shin, FBI: a federated learning-based blockchain-embedded data accumulation scheme using drones for internet of things, IEEE Wirel. Commun. Lett. 11 (5) (2022) 972–976, https://doi.org/10.1109/LWC.2022.3151873.

[29] Yuhao Bai, Qin Hu, Seung-Hyun Seo, Kyubyung Kang, John J. Lee, Public participation consortium blockchain for smart city governance, IEEE Internet Things J. 9 (2020) 2094–2108, https://doi.org/10.1109/JIOT.2021.3091151.

[30] Behnam Khazael, HadiTabatabaee Malazi, Siobhán Clarke, Complex event processing in smart city monitoring applications, IEEE Access 9 (2020) 143150–143165, https://doi.org/10.1109/ACCESS.2021.3119975.

[31] Geetanjali Rathee, Razi Iqbal, Omer Waqar, Ali Kashif Bashir, On the design and implementation of a blockchain enabled E-voting application within IoT-oriented smart cities, IEEE Access 9 (2021) 34165–34176, https://doi.org/10.1109/ACCESS.2021.3061411.

[32] P. McEnroe, S. Wang, M. Liyanage, A survey on the convergence of edge computing and AI for UAVs: opportunities and challenges, IEEE Internet Things J. 9 (17) (2022) 15435–15459, https://doi.org/10.1109/JIOT.2022.3176400.
[33] A. Islam, S. Young Shin, A blockchain-based secure healthcare scheme with the assistance of unmanned aerial vehicle in internet of things, Comput. Electr. Eng. 84 (2020), 106627, https://doi.org/10.1016/j.compeleceng.2020.106627.
[34] H. Wang, et al., Architectural design alternatives based on cloud/edge/fog computing for connected vehicles, IEEE Commun. Surv. Tutor. 22 (4) (2020) 2349–2377, https://doi.org/10.1109/COMST.2020.3020854.
[35] H.A. Alharbi, M. Aldossary, Energy-efficient edge-fog-cloud architecture for iot-based smart agriculture environment, IEEE Access 9 (2021) 110480–110492, https://doi.org/10.1109/ACCESS.2021.3101397.
[36] E. Ahvar, A.-C. Orgerie, A. Lebre, Estimating energy consumption of cloud, fog, and edge computing infrastructures, IEEE Trans. Sustain. Comput. 7 (2) (2022) 277–288, https://doi.org/10.1109/TSUSC.2019.2905900.
[37] A. Islam, M.B. Uddin, M.F. Kader, S.Y. Shin, Blockchain based secure data handover scheme in non-orthogonal multiple access, in: Proceedings of the 2018 4th International Conference on Wireless and Telematics (ICWT), 2018, pp. 1–5. ⟨https://doi.org/10.1109/ICWT.2018.8527732⟩.
[38] J. Kim, et al., A formally verified security scheme for inter-gNB-DU handover in 5G vehicle-to-everything, IEEE Access 9 (2021) 119100–119117, https://doi.org/10.1109/ACCESS.2021.3107308.
[39] B. Wang, M. Li, X. Jin, C. Guo, A reliable IoT edge computing trust management mechanism for smart cities, IEEE Access 8 (2020) 46373–46399, https://doi.org/10.1109/ACCESS.2020.2979022.
[40] N.D. Han, Y. Chung, M. Jo, Green data centers for cloud-assisted mobile ad hoc networks in 5G, IEEE Netw. 29 (2) (2015) 70–76, https://doi.org/10.1109/MNET.2015.7064906.
[41] Q. Li, Y. Zhang, Y. Li, Y. Xiao, X. Ge, Capacity-aware edge caching in fog computing networks, IEEE Trans. Veh. Technol. 69 (8) (2020) 9244–9248, https://doi.org/10.1109/TVT.2020.3001301.
[42] H. Zhao, Q. Wang, J. Wang, B. Wan, Z. Wu, Popularity-based and version-aware caching scheme at edge servers for multi-version VoD systems, IEEE Trans. Circuits Syst. Video Technol. 31 (3) (2021) 1234–1248, https://doi.org/10.1109/TCSVT.2020.2991408.
[43] J. Li, X. Li, Y. Gao, Y. Gao, R. Zhang, Dynamic cloudlet-assisted energy-saving routing mechanism for mobile ad hoc networks, IEEE Access 5 (2017) 20908–20920, https://doi.org/10.1109/ACCESS.2017.2759138.
[44] A. Galletta, M. Fazio, A. Celesti, M. Villari, On the applicability of secret share algorithms for osmotic computing, in: Proceedings of the 2020 IEEE Symposium on Computers and Communications (ISCC), 2020, pp. 1–6. ⟨https://doi.org/10.1109/ISCC50000.2020.9219597⟩.
[45] A. Galletta, J. Taheri, M. Fazio, A. Celesti, M. Villari, Overcoming security limitations of secret share techniques: the nested secret share, in: Proceedings of the 2021 IEEE 20th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom), 2021, pp. 289–96. ⟨https://doi.org/10.1109/TrustCom53373.2021.00054⟩.